

Jesús M. Rueda-Becerril | Ph.D.

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Profile

Research Scientist with expertise in scientific machine learning, numerical simulation, and geospatial data analysis. Strong mathematical foundations with a track record of architecting open-source simulation libraries, leading funded research projects, and translating complex physical models into production-quality scientific software. Experienced working across cross-functional research and engineering teams.

Skills

Languages: Python, C/C++, Fortran 95, Rust, R, Java, SQL, Julia, Shell

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

Simulation: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Experience

Independent Contractor, *Mercor*, Jan 2026 — Present.

- Providing deep expertise in Physics and numerical methods to improve LLM models for top AI labs.

Spatial Data Scientist, *TeaWaters*, May 2025 — Jan 2026.

- Led optimization of scientific code for topographic terrain modeling, improving computational performance of elevation derivative calculations
- Engineered Python tools for geospatial data processing and analysis; utilized random forest ML for wetland probability mapping.
- Performed EDA, feature engineering, and geospatial data preparation including multispectral (Sentinel 1/2) processing for ML/AI land-cover models
- Collaborated with software engineers and scientific team to transition prototype simulation tools into production.

Postdoctoral Research Associate, *Rochester Institute of Technology*, Feb 2021 — Apr 2022.

- Led NSF-sponsored project to upgrade PatchworkMHD, a large-scale C-based HPC simulation code for modeling supermassive black hole binaries using state-of-the-art numerical techniques.
- Implemented new physics features (black hole spin) to the simulation library without impacting runtime, demonstrating expertise in large-scale scientific software architecture.
- Designed and ran large-scale computational experiments on the Frontera supercomputer (TACC), benchmarking and identifying performance optimization opportunities in parallel HPC code.
- Supervised a graduate student applying gradient descent optimization to fit Paramo model parameters to blazar spectral energy distributions observed by the Fermi-LAT telescope, demonstrating ML-assisted parameter estimation workflows for complex astrophysical systems.
- Collaborated on multi-institutional HPC simulation project studying binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, *Purdue University*, Oct 2018 – Nov 2020.

- Architected and developed Paramo, an open-source multiscale/multiphysics simulation library in Fortran 95 modeling relativistic particle distributions via the Fokker-Planck equation coupled with radiative transfer across temporal and spatial scales; speedup via OpenMP parallelization; explored GPU acceleration via OpenACC.
- Implemented advanced numerical methods including sophisticated PDE solvers, numerical integration, and particle distribution algorithms with arbitrary shape support, without impacting simulation runtime.
- Designed Python analysis and visualization tools interfacing with the Paramo simulation library, enabling statistical modeling and parameter optimization workflows.
- Served as a Co-I on two NASA Fermi Guest Investigator Program proposals, one granted; managed 3 PhD researchers and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo*, Jan – Sep 2018.

- Contributed to Python pipelines to process 2D simulation images from the GRTrans numerical code: converted outputs to grayscale, normalized pixel intensities, and extracted spatial features to generate training data for an SVM classifier predicting the black hole spin from radio images.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Software Engineer, *Paychex*, Apr 2022 — Jan 2024.

- Developed Python tests and CI pipelines ensuring software quality standards; deployed production software via Jenkins and OpenShift.
- Conducted SQL data preparation, validation, and analysis from Oracle EBS datasets.

Graduate Research Assistant, *Universitat de València*, Oct 2011 – Jul 2017.

- Developed Python scripts to perform exploratory data analysis on multi-wavelength observational datasets from the NASA Fermi-LAT telescope and the Very Large Baseline Array (VLBA), including sky-coordinate-based source cross-matching across catalogs and log-parabola spectral fitting to characterize blazar emission.
- Implemented numerical tools and data handling extensions to C-SPEV for computing both discrete and continuous multi-wavelength spectra from particle distributions with arbitrary shape
- Published 2 first-author papers on numerical simulations of relativistic plasma dynamics

Projects

Blazar Population ML, *Scientific ML (Python, scikit-learn, PyTorch)*, Jan 2026 — Present.

[GitHub](#)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Developing forecasting pipeline for blazar gamma-ray flares using a time series foundation model, fine-tuned with synthetic light curves generated with Tleco.

WindsOfChange, *Open-Source Lagrangian Particle Simulation Library (R/C++)*, Aug 2024 — Present.

[GitHub](#)

- Developing C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrains with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, *Open-Source Physics Simulation Library (Python/Rust)*, Jan 2024 — Sep 2024.

[GitHub](#)

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, *Open-Source Physics Simulation Code (CONAHCYT Funded)*, Oct 2018 — Apr 2022.

[GitHub](#)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

PatchworkMHD, *Large-Scale MHD Simulation Code (NSF Project)*, Feb 2021 — Apr 2022.

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

Education

Ph.D. in Physics, *Universitat de València, Valencia, Spain*, Oct. 2011 – Jul. 2017.

Excellent *Cum Laude*

M.Sc. in Physics, *Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Mexico*, Aug. 2009 – Sep. 2011.

B.Sc. in Physics, *Universidad Autónoma del Estado de México, Toluca, Mexico*, Aug. 2004 – Dec. 2008.

Selected Publications (9 total, 3 in prep., 4 first-author)

- **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, [ApJ](#) **976**, 182, (2024), [arXiv:2405.17581](#).
- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, [MNRAS](#) **513**, 5766–5779, (2022), [arXiv:2201.07790](#).
- **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, [MNRAS](#) **501**, 4092–4102, (2021), [arXiv:2009.02273](#).
- Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, [MNRAS](#) **496**, 974–986, (2020), [arXiv:1910.14049](#).
- **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, [MNRAS](#) **468**, 1169–1182, (2017), [arXiv:1612.06383](#).